

The Vietnam Findex 2025 Dashboard

Connectivity and Financial Inclusion in the Digital Economy

Technique Application Report

Dataset:	Vietnam – The Global Findex Dataset 2025: Connectivity and Financial Inclusion in the Digital Economy
Course:	Data Visualization
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1 Dataset Overview

The Global Findex 2025 is an appropriate dataset for this report because it combines financial access, digital connectivity, payment behavior, saving, borrowing, and resilience indicators in one nationally comparable survey. That combination makes it possible to study financial inclusion as a progression from readiness and access to usage and outcomes rather than as a single account-ownership rate.

The dashboard analyzed in this report uses the Vietnam microdata collected through face-to-face interviews from June 23 to August 31, 2024. The sample covers 1,000 adults aged 15 years and older and applies survey weights so the reported percentages represent Vietnam's adult population. The survey has an approximate design effect of 1.33 and a margin of error of ± 3.6 percentage points.

The report uses the dataset to answer three linked questions:

- Q1.** Where are the main gaps in digital readiness and formal financial access across income, education, gender, and location?
- Q2.** When adults have accounts, do wages, transfers, sales, and merchant payments actually move through digital channels or revert to cash?
- Q3.** How do different inclusion patterns relate to saving, borrowing, and shock-coping strategies across socioeconomic groups?

2 Dashboard Overview

The Vietnam Financial Inclusion Dashboard 2025 is a policy-facing interactive workspace built from 1,000 weighted Global Findex records for Vietnam. It reorganizes the survey into a five-page analytical flow so readers can move from demographic context to digital readiness, account access, transaction behavior, and financial resilience.

Vietnam Financial Inclusion Dashboard 2025

Global Findex Microdata | Weighted survey results

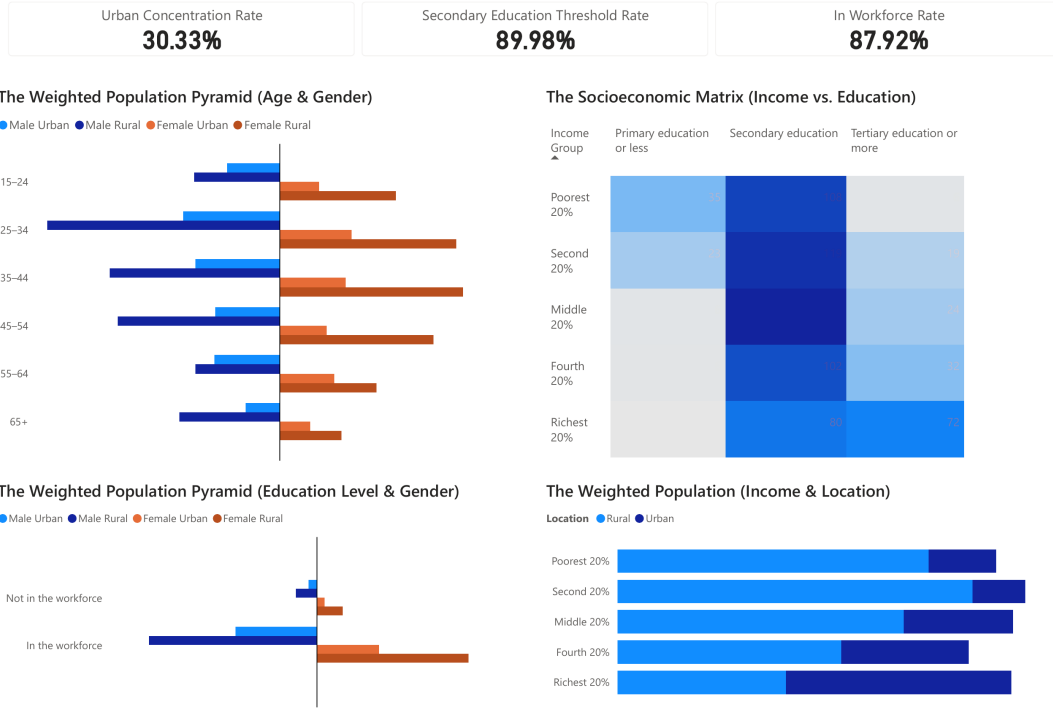


Figure 1: Overview of the Vietnam Financial Inclusion Dashboard.

The page sequence follows a macro-to-micro narrative: Page 1 establishes the demographic baseline, Page 2 checks connectivity and digital identity, Page 3 examines account ownership and exclusion barriers, Page 4 compares cash and digital transaction flows, and Page 5 links financial access with saving, borrowing, and coping capacity. As a result, the dashboard functions as a hybrid explanatory-exploratory system: it presents an authored policy story while still allowing subgroup investigation through interactive filters.

3 Chapter 1: Introduction to Data Visualization

3.1 Techniques / Principles Applied

- High-dimensional data reduction through visual summaries rather than row-by-row table inspection.
- Exploratory and explanatory visualization roles within the same dashboard.
- Policy-facing presentation that highlights actionable disparities instead of isolated statistics.

3.2 How Applied in the Dashboard

The raw dataset in `Findex_Microdata_2025_updateViet Nam.csv` contains 1,000 rows and 199 columns of individual survey information. In raw tabular form, the links between demographics, digital access, payment behavior, and resilience are difficult to see quickly. Visualization addresses that problem by turning survey weights into readable marks such as bar length, color intensity, and flow width, so the dashboard can show where exclusion and cash dependence are concentrated.

The dashboard also combines explanatory and exploratory roles. Its fixed five-page sequence guides the reader from population context to outcomes, while interactive filters on pages such as Account Ownership & Access Landscape and Financial Health & Resilience let analysts inspect specific subgroups. This combination is appropriate for a policy audience because it keeps the national story coherent while still supporting diagnosis.

3.3 Notes / Adjustments

Many visualization guidelines separate exploratory tools from guided presentations, but this dashboard places filters inside an authored narrative. That deviation is justified because financial inclusion questions are strongly intersectional, so readers often need to move from the national story to a specific subgroup without leaving the dashboard.

4 Chapter 2: Data Types, Marks, and Channels

4.1 Techniques / Principles Applied

- Classifying variables as nominal, ordinal, or quantitative before encoding them.
- Selecting marks that fit the analytical task: cells, bars, flows, and lines.
- Prioritizing precise channels such as aligned position and length for the most important quantitative comparisons.
- Using color, width, and enclosure as supporting channels rather than the primary source of exact comparison.

4.2 How Applied in the Dashboard

The dashboard aligns each page with the variable types and comparison task it needs to support. Table 1 summarizes the main encodings used across the five pages.

Chart/Page	Variables	Marks / channels	Why appropriate
Page 1 socioeco- nomic matrix	Income quintile, edu- cation, survey weight	Grid cells; row and col- umn position; sequen- tial color intensity	Two ordered categorical variables form a discrete matrix, while weighted population density is con- veyed by ordered color.
Page 2 mobile ownership pro- file	Location, gender, in- come group, phone type, access rate	Bars; aligned length; grouped position; hue	Bar length supports ac- curate rate comparison across small multiples and hue separates phone types without changing the baseline.
Page 3 account composition flow	Location, account state, weighted vol- ume	Sankey nodes and rib- bons; horizontal po- sition; ribbon width; color grouping	The chart emphasizes how groups move between account states, which is more important here than exact pairwise magnitude reading.
Page 4 inflow digitization	Inflow type, payment modality, percentage share	Segmented bars; aligned position; seg- ment length; hue	A 100% stacked bar makes composition com- parable across wages, agri- culture, transfers, and other inflows.
Page 5 save vs. borrow trajec- tories	Income group, income runway, save rate, bor- row rate	Lines; vertical position; color	The line overlays help readers track directional change across ordered cat- egories.

Table 1: Main marks, channels, and encoding rationale used in the dashboard.

The Socioeconomic Matrix (Income vs. Education)

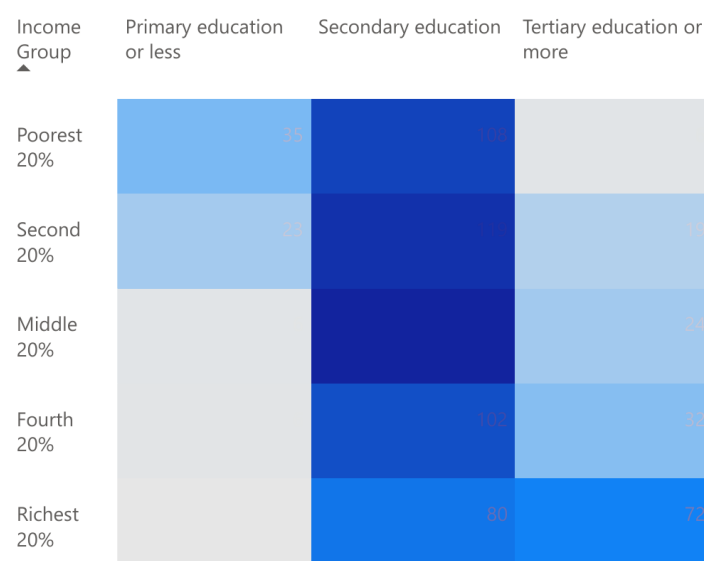


Figure 2: The socioeconomic matrix (income vs. education).

Mobile Ownership % by Location, Phone Type, Gender and Income Group

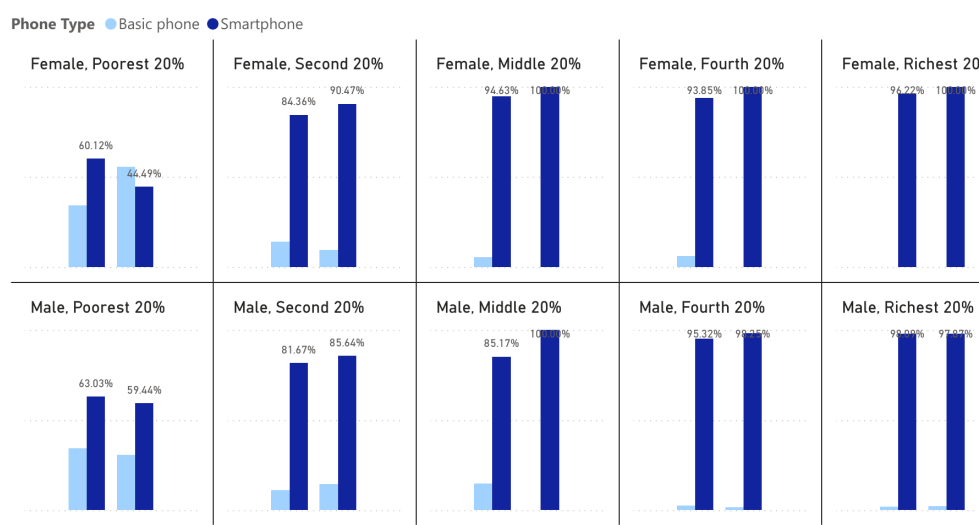


Figure 3: Mobile ownership rate by location, phone type, gender, and income group.

The Account Composition Flow

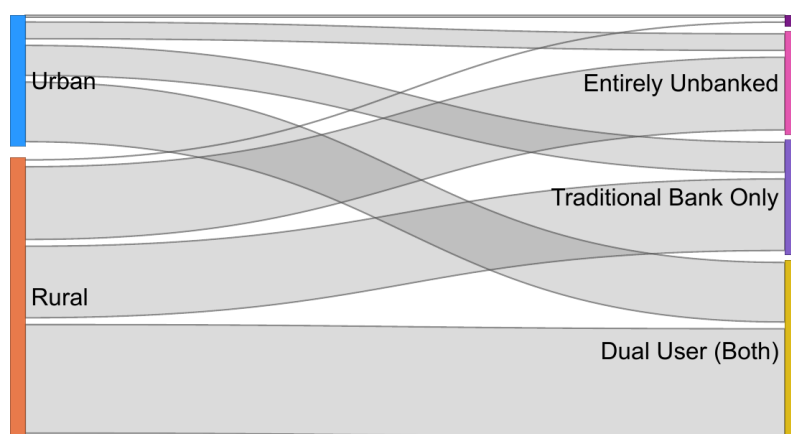


Figure 4: The account composition flow.

National Digitization of Financial Inflows: Cash vs Digital Delivery

Receive Payment Method ● Cash Only ● Digital (Direct to Account) ● Other Methods ● Unknown

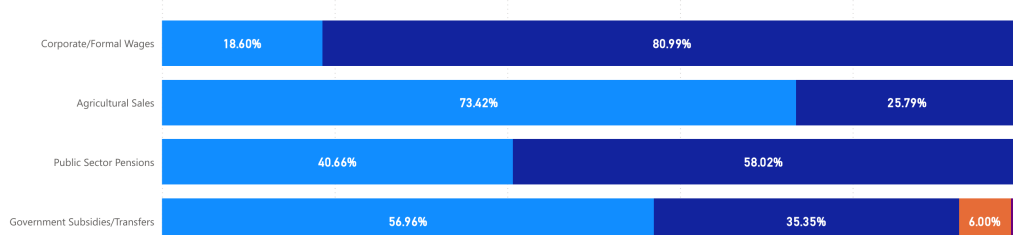
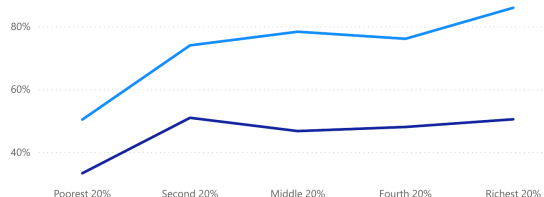


Figure 5: The national digitization of financial inflows: cash vs. digital delivery.

Save vs. Borrow Percentage by Income Group

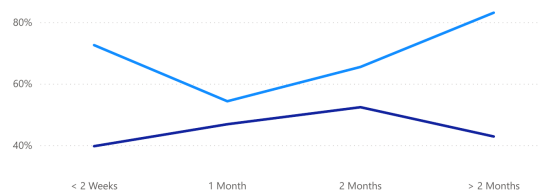
● % Who Saved ● % Who Borrowed



(a) By income group.

Save vs. Borrow Percentage by Income Runway Length

● % Who Saved ● % Who Borrowed



(b) By income runway length.

Figure 6: Save vs. borrow percentage overlays.

These examples show the dashboard's encoding logic in practice: Page 1 uses a matrix to compare ordered categories, Page 2 uses small-multiples bars for precise subgroup comparisons, Page 3 uses a Sankey to emphasize structural flows, Page 4 uses segmented bars for composition, and Page 5 uses line overlays to highlight directional change across ordered groups.

4.3 Notes / Adjustments

The Sankey on Page 3 sacrifices some magnitude precision because ribbon width is less exact than bar length, but it better communicates movement between account states. On Page 5, the line overlays connect ordinal categories, so they imply continuity more than a strict categorical display would; the choice is kept to make change in save-versus-borrow behavior easier to track.

5 Chapter 3: Pre-Attentive Processing, Gestalt Principles, and Magnitude Estimation

5.1 Techniques / Principles Applied

- Pre-attentive processing through contrast, saturation, and enclosure.
- Gestalt grouping, especially proximity, similarity, and enclosure.
- Magnitude estimation that favors common baselines and linear length for important comparisons.

5.2 How Applied in the Dashboard

Pre-attentive pop-out is used to direct attention before detailed reading. On Page 1, Figure 2 uses darker blue cells to make high-density income–education intersections stand out quickly. On Page 2, the mobile ownership chart differentiates smartphone and basic phone rates with strong luminance contrast, so the penetration gap is visible without a full label-by-label scan.

Banking Inclusion % 74.93%	Financial Institution Account % 74.05%	Mobile Money Account % 46.69%	Digital Account % 67.03%
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Figure 7: The Account Ownership & Access Landscape KPI cards.

Gestalt grouping is especially visible on Page 3. Figure 7 clusters the four KPI cards closely together, so proximity signals that they belong to the same summary layer. Enclosure appears in the filter strip, where the controls are visually separated from the charts they drive.

FILTER	Urbanicity	Gender	In Workforce	Education Level	Income Group
	All ▾	All ▾	All ▾	All ▾	All ▾

Figure 8: Filter controls on the Account Ownership & Access Landscape page.

Magnitude estimation is handled conservatively in the comparison-heavy charts. The Mobile Ownership Profile and the Exclusion Friction Log rely on bar length and aligned baselines because those channels are more accurate for comparing rates than area- or angle-based alternatives.

5.3 Notes / Adjustments

The Account Composition Flow in Figure 4 is less precise than a standard bar chart for direct magnitude comparison. The trade-off is accepted because the page needs to emphasize relationships between source groups and account states, not only isolated totals.

6 Chapter 4: Chart Types for Amounts, Distributions, Proportions, Relationships, and Trends

6.1 Techniques / Principles Applied

- Matching chart types to amounts, distributions, proportions, relationships, and trends.
- Preferring readable 2D Cartesian layouts for most dashboard comparisons.
- Recognizing when uncertainty is not displayed and what that implies for interpretation.

6.2 How Applied in the Dashboard

Why Consumers Stick to Cash-Only Transactions

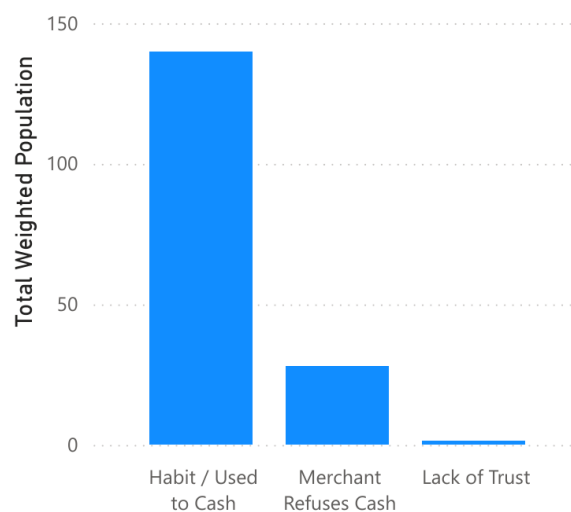


Figure 9: Why consumers stick to cash-only transactions.

Amounts. Figure 9 uses a standard bar chart to compare the weighted size of cash-only barriers. This is appropriate because the task is simple magnitude ranking, and bar length on a common baseline makes the differences easy to read.

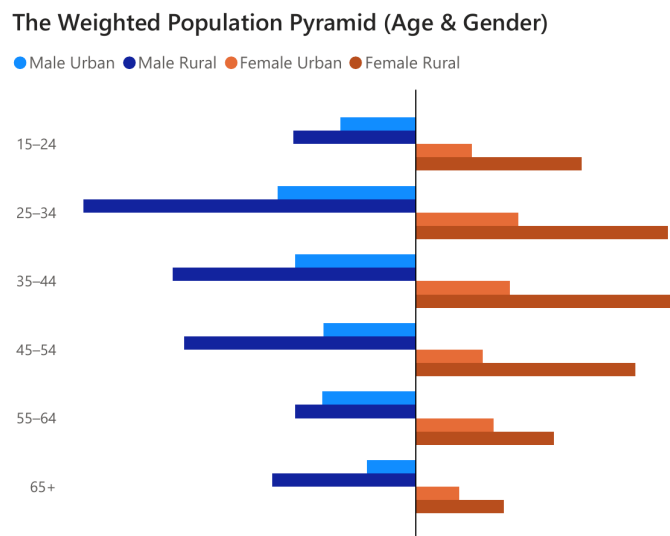


Figure 10: The weighted population pyramid by age and gender.

Distributions. Figure 10 and the socioeconomic matrix show how the population is distributed across age, gender, income, and education. These views are suitable because they reveal whether the sample is balanced or concentrated in specific groups before later policy comparisons begin.

Proportions. Figure 5 uses a 100% stacked bar layout to compare payment composition within each inflow type. This chart works well when the goal is to compare how cash, direct-to-account, and other channels share the same total.

Relationships. Figure 3 uses a small-multiples bar design to compare ownership patterns across location, gender, income, and phone type. The structure avoids overplotting while still showing how the variables relate to one another.

Trends. Figure 6 uses line overlays to show how saving and borrowing rates move across ordered income groups and income runway categories. The chart is effective for showing direction of change even though the categories are ordinal rather than continuous.

Across these examples, the dashboard relies mainly on a 2D Cartesian coordinate system because perpendicular axes support clearer comparison than radial or 3D alternatives for this type of policy dashboard. Statistical uncertainty is not explicitly visualized, so the charts are easy to read but can overstate precision, especially for filtered subgroup views.

6.3 Notes / Adjustments

The 100% stacked inflow chart emphasizes composition rather than exact comparison of every internal segment, so only the first segment in each row has a shared baseline. The

save-versus-borrow lines also simplify ordinal data into continuous-looking trajectories, which improves readability but should be interpreted as directional rather than strictly continuous change.

7 Chapter 6: Visual Integrity (Ink, Overlap, Color, Layout, and Annotation)

7.1 Techniques / Principles Applied

- Proportional ink and true baselines for quantitative marks.
- Color choices that preserve contrast and basic accessibility.
- Overlap control, annotation, and data–context balance.

7.2 How Applied in the Dashboard

The dashboard maintains proportional ink by anchoring bar charts at true zero baselines and by keeping bar length or flow width tied to the underlying weighted values. This is visible in Figure 9 and in the previously introduced account-composition Sankey.

Color is used with restraint. Rural and urban comparisons rely on strong luminance contrast, Page 3 gender comparisons avoid low-contrast pairings, and the socioeconomic matrix keeps a single sequential palette so density is shown through intensity rather than extra hue variation.

Overlap is handled through panel separation, spacing, and concise annotation. Dense views are broken into smaller comparison units, while reference titles, legends, and benchmark markers provide context without covering the data.

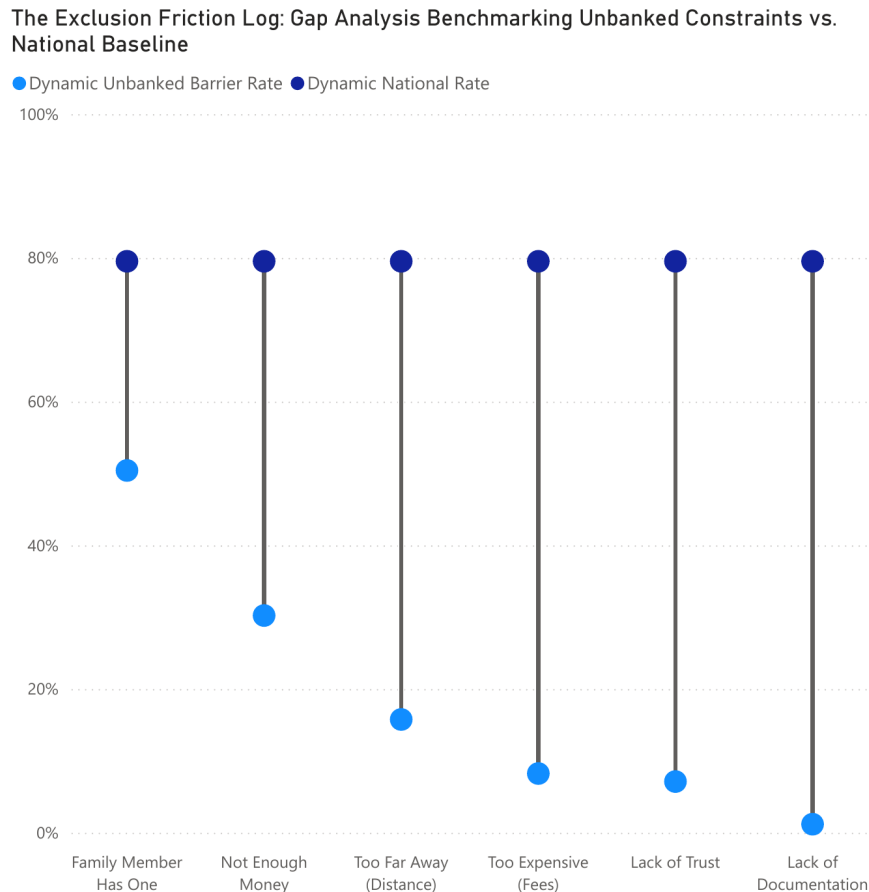


Figure 11: The Exclusion Friction Log: gap analysis benchmarking unbanked constraints vs. national baseline.

Figure 11 is a good example of data–context balance because the title, legend, and reference markers explain the benchmark logic without burying the chart under decorative elements.

7.3 Notes / Adjustments

The National Digitization of Financial Inflows chart in Figure 5 uses a 100% stacked format, so only the first segment in each row has a common baseline. That reduces precision for the inner segments, but the compromise is acceptable because the page is intended to show composition rather than exact segment-by-segment ranking.

8 Chapter 8: Interaction Techniques

8.1 Techniques / Principles Applied

- Overview first, then filter, then details-on-demand.
- Coordinated views that update several visuals from the same selection.

- Local page-level filters instead of one global interaction state.
- Power BI slicers and linked visuals as the main dashboard interaction tools.

8.2 How Applied in the Dashboard

The dashboard was built in Power BI and uses interaction mainly to extend a guided policy narrative into subgroup exploration. Users first see overview states on pages such as Account Ownership & Access Landscape and Financial Health & Resilience, then narrow the analysis through slicers, and finally use details-on-demand in the live dashboard to inspect exact values.



Figure 12: List of filters on the Account Ownership & Access Landscape page.

Figure 12 shows the main interaction design on Page 3: the page starts with a national overview, then lets users filter by urbanicity, income group, gender, education level, and workforce status. Within the live Power BI dashboard, those slicers coordinate the KPI cards, the Sankey diagram, and the exclusion-barrier chart so that all three reflect the same selected cohort.

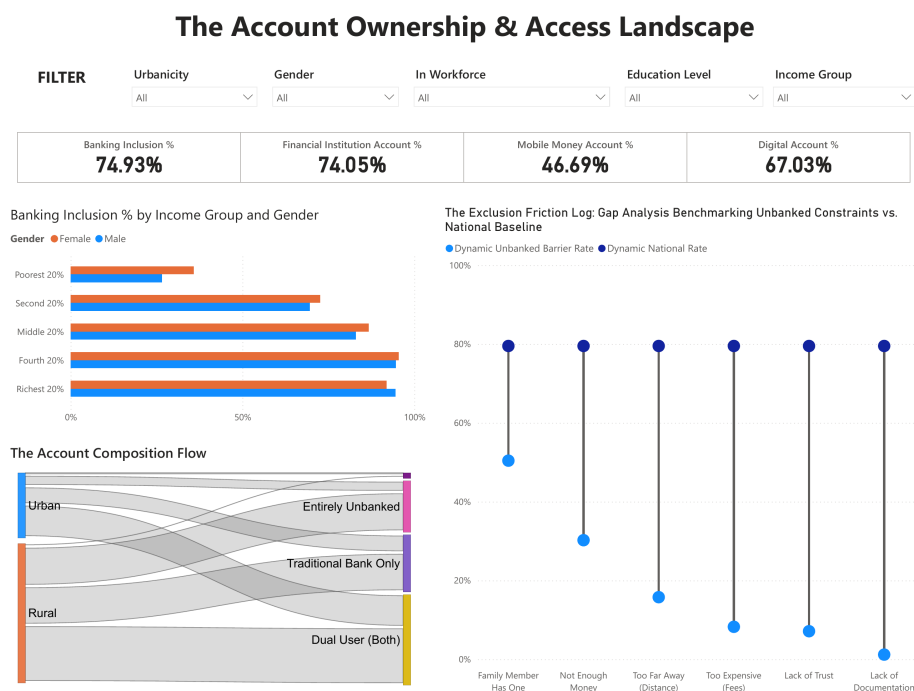


Figure 13: Default Account Ownership & Access Landscape state visible in the static PDF export.

The exported PDF confirms the placement of the filters and the default overview state. It also supports the claim that the dashboard uses local page-level interaction rather than one global control state shared across every page.

8.3 Notes / Adjustments

Because the available source is a static PDF export, this report cannot verify live hover tooltips, zoom behavior, animation, or the exact values produced by every filtered state. The chapter therefore describes the dashboard as designed for filtering, coordinated updates, and details-on-demand without claiming interactive evidence that is not directly visible in the exported file.

9 Chapter 9: Storytelling and Presentation Narrative

This chapter reads the Vietnam Financial Inclusion Dashboard 2025 as a policy brief built from weighted survey evidence rather than as a loose collection of charts. Its central argument is that financial inclusion in Vietnam cannot be judged by account ownership alone; it must be read as a progression from social structure to digital readiness, from access to actual usage, and finally to financial resilience.

See Figure 1.

9.1 The Core Policy Story

The chapter follows one narrative spine only:

Context → Readiness → Access → Usage → Resilience

This sequence is persuasive because each page answers the most natural policy question created by the page before it. A country can appear digitally connected yet still fail to activate digital identity. It can appear broadly banked yet still settle routine transactions in cash. It can appear financially included yet still leave vulnerable households dependent on informal support when shocks arrive.

**Context → Readiness → Access → Usage →
Resilience**

Figure 14: Narrative roadmap of the Vietnam Financial Inclusion Dashboard.

9.1.1 Page 1: Establishing the Structural Baseline

Page 1 serves as the structural baseline of the story, showing that later gaps in access and usage emerge inside an already uneven social landscape rather than in a neutral population.

The strongest evidence comes from the macro indicators and the socioeconomic matrix in Figure 2. The page reports an Urban Concentration Rate of only 30.33%, an In Workforce Rate of 87.92%, and a Secondary Education Threshold Rate of 89.98%. Yet the matrix shows that the Poorest 20% group has only 6 respondents with tertiary education or more, compared with 72 in the Richest 20% group. Those numbers make inequality in income and education visible before the dashboard ever begins talking about banking.

That opening matters for the policy argument because it prevents the reader from treating later financial outcomes as purely technological problems. The story begins by showing that exclusion is already structured by geography, socioeconomic position, and educational opportunity, so any later digital or financial intervention must be judged against that baseline rather than against national averages alone.

See Figure 2.

9.1.2 Page 2: Readiness Exists, but Activation Is Weaker

Page 2 advances the story by testing whether the population that looks demographically ready for participation is also ready to use digital financial tools in practice.

Figure 15 shows that internet access is both high and frequent: Internet Use in the last three months reaches 87.85%, while Internet Use in the last seven days remains close at 85.39%. The same figure also shows a sharp activation gap: Foundational ID coverage reaches 99.65%, Online Digital ID reaches 71.19%, but Active Digital ID Verification falls to only 32.68%. Figure 16 pushes the argument further: Social Media reaches 94.46% and Read News reaches 78.44%, but Gov activity is only 15.63% and Job activity is only 4.89%.

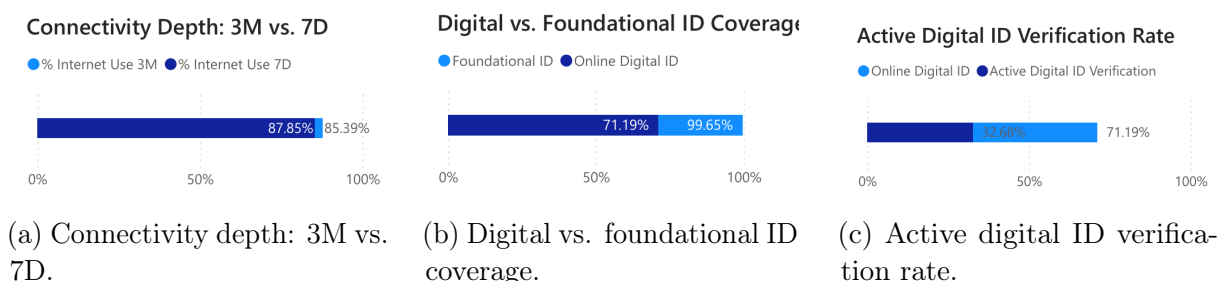


Figure 15: Page 2 readiness evidence used in the storytelling analysis.

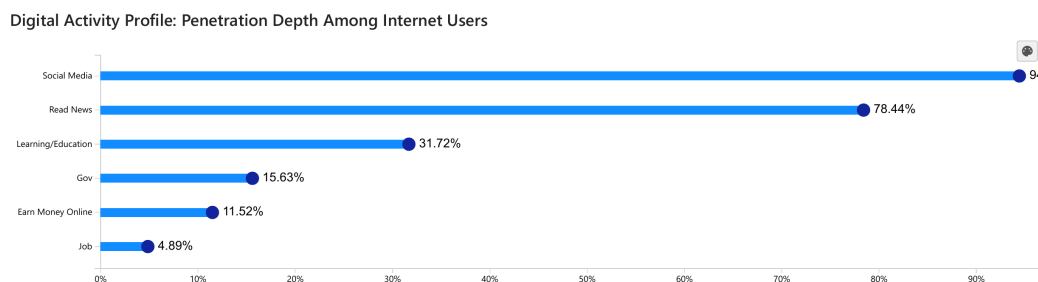


Figure 16: Digital activity profile: penetration depth among internet users.

Together, these figures argue that connectivity alone is not the same as activation. Vietnam's adults appear highly present online, but their digital behavior remains much stronger in communication and information than in institutional or economic use. For a policy reader, this reframes the problem: the bottleneck is not simply device or internet availability, but the conversion of digital presence into verified, trusted, high-value use.

9.1.3 Page 3: Access Is Broad, but Not Even

Page 3 becomes the diagnostic center of the story by asking who is formally included, who is left behind, and what kind of inclusion the system is actually delivering.

The KPI cards in Figure 7 report Banking Inclusion at 74.93%, Financial Institution Account ownership at 74.05%, Mobile Money Account ownership at 46.69%, and Digital Account ownership at 67.03%. That already shows a 27.36 percentage-point gap between financial institution accounts and mobile money accounts. Figure 17 and Figure 4 then show that access is not socially neutral: lower-income groups, especially on the rural side of the account-composition flow, are more exposed to unbanked or thinner account states. Figure 11 adds a barrier layer to that story by showing that exclusion is not just a missing account status but a set of recurring frictions tied to money, cost, and access conditions.

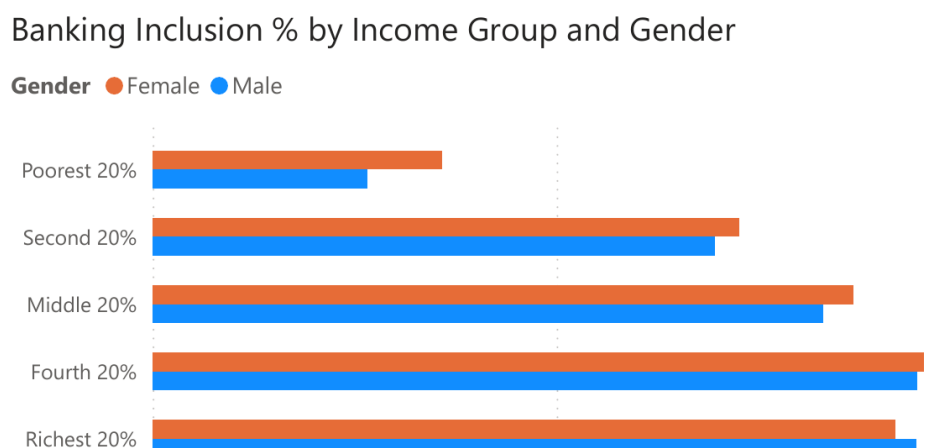


Figure 17: Banking inclusion by income group and gender.

See also Figures 7, 8, 4, and 11.

The policy consequence is that high national access cannot be read as uniform inclusion. A dashboard that stopped at the 74.93% headline would suggest a largely solved problem; this page instead shows that the remaining gap is concentrated, structured, and diagnostically meaningful. The story therefore shifts from “how many people are banked?” to “which groups remain thinly served, and what keeps them there?”

9.1.4 Page 4: Access Does Not Guarantee Digital Usage

Page 4 challenges the access story by asking whether accounts actually change how money moves through everyday life.

Figure 5 shows that Corporate/Formal Wages are mostly digitized, with 80.99% delivered directly to an account and only 18.60% cash-only. Agricultural Sales show the opposite pattern: 73.42% are cash-only and only 25.79% are delivered digitally to an account. Figure 18 adds a geographic gap: Urban Merchant Payments reach 65.79%, compared with 50.30% in rural areas, while Urban In-Store Digital Pay reaches 62.67%, compared with 46.11% in rural areas. Figure 19 visually reinforces the same divide by showing that rural usage leans more heavily toward cash-only or non-bill-paying states than the urban profile.

Digital Merchant and In-Store Payment

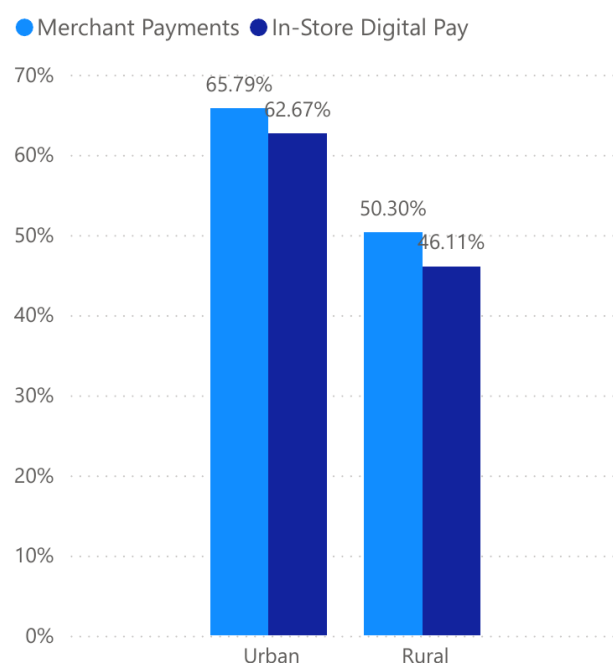


Figure 18: Digital merchant and in-store payment by geography.

See Figure 5.

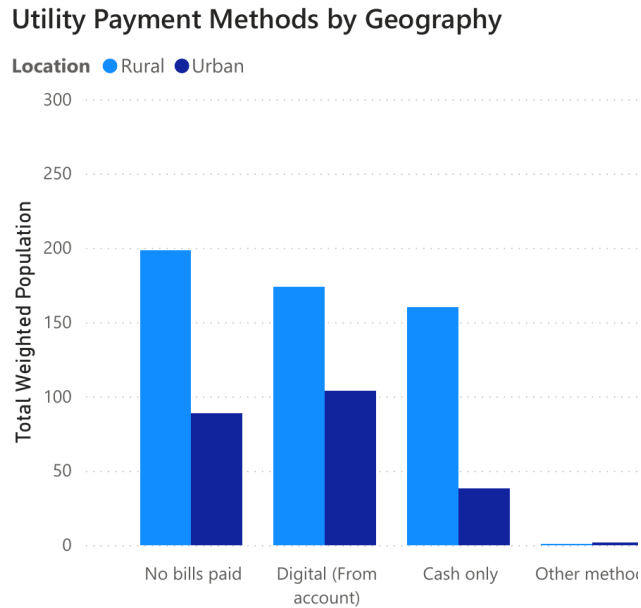


Figure 19: Utility payment methods by geography.

This is the chapter’s most important argumentative turn. Page 3 could have been read as a success story about account ownership; Page 4 shows why that reading is incomplete. The policy issue is not only whether people possess accounts, but whether rural payment ecosystems, merchants, and routine household transactions are digitized strongly enough for those accounts to matter in practice.

9.1.5 Page 5: The Real Test Is Resilience Under Stress

Page 5 gives the story its outcome test by asking whether participation in the financial system is associated with stronger household resilience rather than with access alone.

Figure 20 shows a national macro split in behavior: 73.05% of respondents saved, while 45.98% borrowed. That national picture is not the endpoint, however. Figure 6 shows that saving and borrowing move differently across income groups and income runway lengths, while Figure 22 reveals a much sharper difference in coping channels. In the Poorest 20% group, Family & Friends stands at 86 while Formal Savings is only 30; in the Richest 20% group, Formal Savings rises to 60 while Family & Friends falls to 34. The full page overview in Figure 21 shows how these KPI, trajectory, and coping views work together inside one resilience frame.



Figure 20: Page 5 resilience KPI cards.

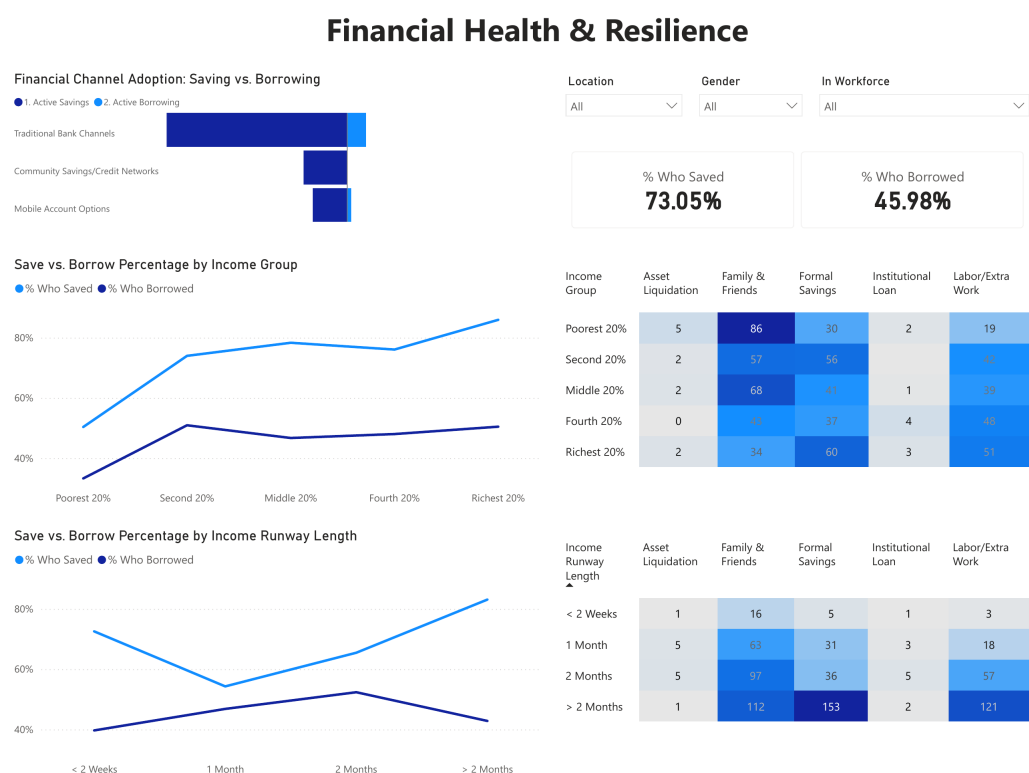


Figure 21: Financial Health & Resilience page overview.

Income Group	Asset Liquidation	Family & Friends	Formal Savings	Institutional Loan	Labor/Extra Work
Poorest 20%	5	86	30	2	19
Second 20%	2	57	56		42
Middle 20%	2	68	41	1	39
Fourth 20%	0	43	37	4	48
Richest 20%	2	34	60	3	51

Figure 22: Coping channels by income group.

The final implication is that inclusion should be judged by how households absorb shocks, not only by whether they hold an account. The dashboard ends on resilience because it reveals a substantive difference between formal financial participation and fallback dependence on informal networks. That ending gives the chapter its human and policy-facing conclusion: the strongest financial systems are not just those that open accounts, but those that reduce vulnerability when households are under pressure.

9.2 Why the Sequence Persuades

The sequence persuades because it prevents the reader from jumping too quickly to a positive conclusion about inclusion. The chapter begins with structural conditions, moves through readiness and access, tests whether those conditions produce real digital usage, and only then asks whether that usage translates into resilience. The same story can still be summarized once as *Context* → *Readiness* → *Access* → *Usage* → *Resilience*, but the force of the sequence lies in what each transition corrects.

The key corrective turn happens after Page 3. Banking Inclusion reaches 74.93%, and Financial Institution Account ownership reaches 74.05%, which could easily support a success narrative. Page 4 interrupts that reading by showing Agricultural Sales at 73.42% cash-only and Urban Merchant Payments at 65.79% versus 50.30% in rural areas. Page 5 then interrupts any remaining optimism by showing that the Poorest 20% still rely far more on Family & Friends at 86 than on Formal Savings at 30. In other words, the chapter persuades because it steadily narrows the distance between headline access and lived financial security.

See Figures 14, 7, 5, and 22.

9.3 How the Evidence Builds the Argument

The argument is built through accumulation rather than assertion. Each page begins with a high-level anchor, then moves into a more diagnostic chart that either qualifies or complicates the headline. For example, Page 2 does not stop at high internet-use rates of 87.85% and 85.39%; it contrasts them with Active Digital ID Verification at only 32.68%, then deepens the point through Figure 16, where Social Media reaches 94.46% but Gov activity is only 15.63%. That sequence turns a connectivity story into an activation story.

The same evidence-building logic continues in later pages. Page 3 moves from the 74.93% banking headline to social segmentation and exclusion barriers. Page 4 moves from digital wage inflows at 80.99% to the much weaker rural and agricultural usage environment. Page 5 moves from a national Saved rate of 73.05% to the more revealing contrast between Family & Friends at 86 and Formal Savings at 30 for the Poorest 20%. Because each visual layer narrows the meaning of the previous one, the reader is led toward a more cautious and policy-useful interpretation of inclusion.

See Figures 15, 16, 7, 19, and 22.

9.4 Narrative Style and Claim Discipline

The narrative style is persuasive, but it stays disciplined by grounding every major interpretation in visible comparison rather than in speculative language. The chapter does not claim that digital access *causes* resilience or that cash usage *proves* exclusion. Instead,

it repeatedly uses measured gaps to show where the financial system appears stronger, where it appears weaker, and which groups or settings deserve closer policy attention. That is why the tone works better as a policy brief than as a promotional story.

This discipline matters most where the chapter could easily overstate its case. The gap between Online Digital ID at 71.19% and Active Digital ID Verification at 32.68% supports a claim about weaker activation, not a claim that connectivity has failed altogether. The contrast between Formal Wages at 80.99% direct-to-account and Agricultural Sales at 73.42% cash-only supports a claim about uneven digitization, not a claim that all rural finance is static. Likewise, the Poorest 20% reliance on Family & Friends at 86 versus Formal Savings at 30 supports a claim about vulnerability, not a deterministic claim about all low-income households. The chapter is most credible when it keeps that distinction clear.

See Figures 16, 5, 18, and 20.

9.5 Interaction as Policy Exploration

The chapter's policy argument is strengthened by interaction because the dashboard does not force the reader to stop at the national averages used in the narrative. Page 3 begins with Banking Inclusion at 74.93%, Financial Institution Account ownership at 74.05%, Mobile Money Account ownership at 46.69%, and Digital Account ownership at 67.03%, then gives the user filters for Urbanicity, Income Group, Gender, Education Level, and In Workforce. Page 5 does something similar from a resilience baseline of 73.05% saved and 45.98% borrowed, using filters for Location, Gender, and In Workforce to move the reader from national averages into subgroup patterns.



Location	Gender	In Workforce
All	All	All

Figure 23: Filter controls on the Financial Health & Resilience page.

This interaction design matters because the chapter's most important claims are distributional. The national story says that digital activation is weaker than connectivity, that account access is broader than digital usage, and that resilience differs across social groups. The filters on Page 3 and Page 5 make those claims investigable instead of merely rhetorical. At the same time, the available source is still a static PDF export, so the report should only claim that the dashboard is *designed for* subgroup exploration, not that every filtered result has been directly tested from the exported file.

See Figures 8, 23, 13, and 21.